

UNIVERSITY OF SURREY ©

Faculty of Engineering and Physical Sciences
School of Mathematics and Physics
Department of Mathematics

Undergraduate Programmes in Mathematics

Module MAT3032; 15 Credits

Advanced Algebra

FHEQ Level 6 (Year 3) Examination

Time allowed: **2 hours**

Semester 1 2023/4

Answer **THREE** questions only. If you attempt more than three questions, then only the BEST THREE solutions will be taken into account. Show full workings.

Results proved in the module may be assumed and used without proof, unless a proof is requested.

Where appropriate, the mark carried by an individual part of a question is indicated in square brackets [], and the maximum mark of this assessment is 75.

Candidates may only use calculators that are non-programmable, with no alphanumeric memory and not wireless enabled.

Additional Material:

one side of A4 paper of handwritten and/or typeset notes, 0 handouts

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Question 1

- (a) Let G be a finite multiplicative group with identity element e . Suppose H is an abelian subgroup of G . For any $g \in G$ and $h \in H$ let $h(g) = hgh^3$.

(i) Show the rule $(h, g) \mapsto h(g)$ defines a group action of H on G . **[6 marks]**

Suppose, in addition, that $h^4 = e$ for all $h \in H$.

(ii) Show that for all $g \in G$ the orbit $\mathcal{O}(g)$ is contained in the conjugacy class of $g \in G$. **[4 marks]**

(iii) Show that $H \subset \text{Fix}(H)$, where $\text{Fix}(H)$ is the fixed set of H . **[4 marks]**

- (b) Consider the group $(\mathbb{Z}_{20}, +_{20})$ of integers modulo 20 with addition modulo 20.

(i) Find the number of automorphisms of $(\mathbb{Z}_{20}, +_{20})$ and describe *one* of the automorphisms by its effect on the generator 1 of $(\mathbb{Z}_{20}, +_{20})$. **[4 marks]**

(ii) State, with a reason, how many automorphisms of $(\mathbb{Z}_{20}, +_{20})$ are inner automorphisms. **[2 marks]**

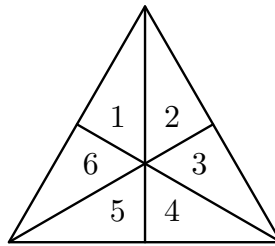
- (c) Suppose H is a subgroup of index m in a finite group G . Show that G has a normal subgroup N which is contained in G such that $[G : N] \leq m!$. **[5 marks]**

[Total for Question 1: 25 marks]

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Question 2

- (a) An equilateral triangle, made out of transparent material, is divided into six congruent regions as shown. Each region is to be coloured with one of q colours, each of which may be used any number of times. The triangle may be rotated in its plane and/or turned over.



- (i) Express all the symmetries of the figure as permutations of $\{1, 2, 3, 4, 5, 6\}$; each permutation should be expressed as a product of disjoint cycles. **[6 marks]**
 - (ii) Give a presentation of the symmetry group of the figure and identify a group the symmetry group is isomorphic to. **[5 marks]**
 - (iii) Find the number of distinguishable q -colourings of the figure. **[4 marks]**
- (b) Let $\sigma = (1 \ 2)(4 \ 5)(3) \in S_5$, where (3) represents the 1-cycle $3 \mapsto 3$.
- (i) State the order of σ in S_5 . **[2 marks]**
 - (ii) State the cycle type and index of σ and find expressions for the number of elements in the centraliser and conjugacy class of σ in S_5 . **[4 marks]**
 - (iii) Find the number of elements in S_5 with the same order as σ . **[4 marks]**

[Total for Question 2: 25 marks]

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Question 3

- (a) Let G be a group of order $520 = 2^3 \times 5 \times 13$.
- (i) For each prime factor p of $|G|$, find the possible numbers of p -Sylow subgroups of G and state the orders of these subgroups. **[7 marks]**
 - (ii) By considering the number of 5-Sylows and the number of 13-Sylows in G , show G is not simple. **[7 marks]**
 - (iii) Suppose G has one normal subgroup N of order 13 and consider G/N . Observe $|G/N| = 40 = 2^3 \times 5$. Find the number of 5-Sylows in G/N . Show that G has a normal subgroup of order 65. **[5 marks]**
- (b) Consider a group G of order $1001 = 7 \times 11 \times 13$ with a subgroup H of order $143 = 11 \times 13$.
- (i) Show that, up to isomorphism, there is only one group of order 143, namely the cyclic group of order 143. **[3 marks]**
 - (ii) Show that H is a normal subgroup of G and deduce $H \subset Z(G)$ where $Z(G)$ is the centre of G . **[3 marks]**

[Total for Question 3: 25 marks]

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Question 4

(a) Let $\mathbb{H}$ be the division algebra of real quaternions with basis $\{1, i, j, k\}$. Recall that $i^2 = j^2 = k^2 = ijk = -1$ and $ij = -ji = k, jk = -kj = i$ and $ki = -ik = j$.

(i) Use the quaternions $2 + 4i + j + 2k$ and $3 + 2i + j + 3k$ to find positive integers w, x, y and z such that $(2^2 + 4^2 + 1^2 + 2^2)(3^2 + 2^2 + 1^2 + 3^2) = w^2 + x^2 + y^2 + z^2$.

[6 marks]

(ii) Recall that $\mathbb{H}$ over $\mathbb{R}$ is an $\mathbb{R}$ -algebra. Find the matrix which represents $2 + 4i + j + 2k$ in the regular representation of $\mathbb{H}$ relative to $(1, i, j, k)$.

[6 marks]

(iii) Suppose u and v are quaternions such that $v^2 = u$. Show that $(u - v(v + \bar{v}))^2 = \|u\|^2$, where $\|u\|$ is the norm of u . Hence find the two possible square roots of $2 + 4i + j + 2k$ in $\mathbb{H}$. (You may find it useful to recall that $\|w^2\| = \|w\|^2$ for any quaternion w .)

[5 marks]

(b) Suppose e and f are non-trivial idempotent elements of a ring R such that $ef = fe$. Let $g = e + f - ef$.

(i) Show g is an idempotent element of R .

[4 marks]

(ii) Show that $g = r_1e + r_2f$ where $r_1 = e$ and $r_2 = f - e$. Hence explain why $Rg \subset Re + Rf$. Furthermore, show that $eg = e$ and $fg = f$. Deduce $Rg = Re + Rf$.

[4 marks]

[Total for Question 4: 25 marks]